

# Chapter 27 Circuits

## Resistors in Series and Parallel

### Resistors in Series

$$\begin{aligned}I_1 &= I_2 = \dots = I \\V_1 + V_2 + \dots &= V \\R_{eq} &= R_1 + R_2 + \dots\end{aligned}$$

### Resistors in Parallel

$$\begin{aligned}I_1 + I_2 + \dots &= I \\V_1 &= V_2 = \dots = V \\\frac{1}{R_{eq}} &= \frac{1}{R_1} + \frac{1}{R_2} + \dots\end{aligned}$$

## Kirchhoff's Rules

Some definition:

- Junction: Connection point of **three or more wires**
- Loop: Any closed circuit in a circuit
- Sign conventions:
  - The current flowing **into the junction is positive** and **out is negative**.
  - The direction in which the circuit circulates is **firstly defined**: along this direction, voltage rise is positive, drop is negative.

(a) Sign conventions for emfs

$+\mathcal{E}$ : Travel direction  
from  $-$  to  $+$ :



$-\mathcal{E}$ : Travel direction  
from  $+$  to  $-$ :

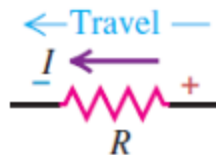


(b) Sign conventions for resistors

$+IR$ : Travel *opposite*  
to current direction:



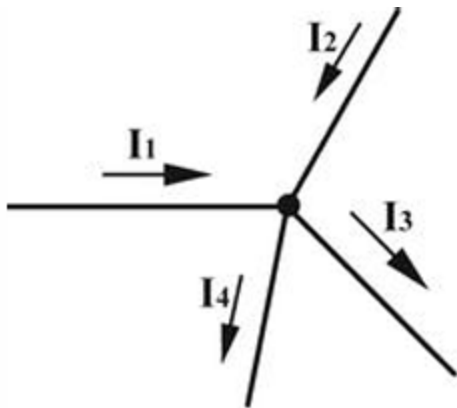
$-IR$ : Travel *in*  
current direction:



## Kirchhoff's Junction Rule (Conservation of Charge)

The **algebraic sum** of all current intensities at the input and output junction is zero.

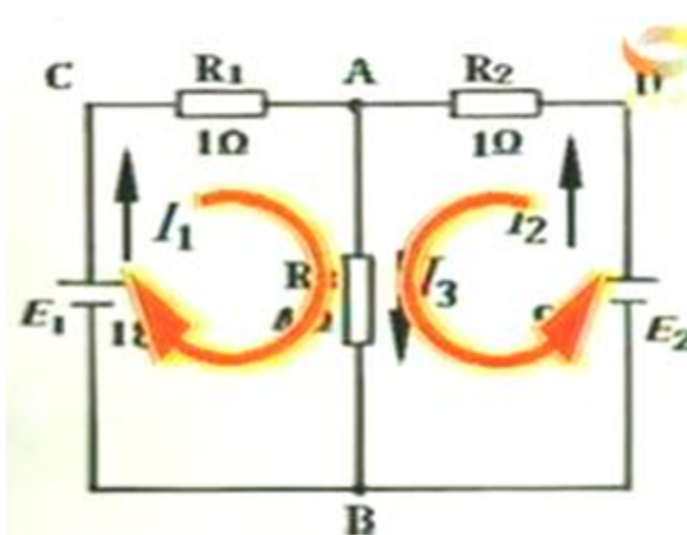
$$\sum I_i = 0$$



## Kirchhoff's Loop Rule (Conservation of Energy)

The algebraic sum of all voltages along any loop is zero.

$$\sum V_i = 0$$



## Problem-Solving Strategy

- Draw the circuit diagram and assign labels and symbols to all known and unknown quantities.
- Assign directions to the currents.
- Apply the junction rule to any junction in the circuit.
- Apply the loop rule to needed loops
- Solve the equations simultaneously for the unknown quantities.

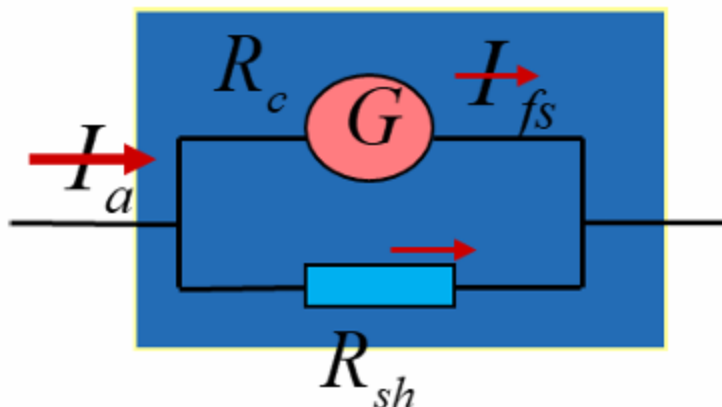
## Electrical Measuring Instruments

Commonly Used instruments: ammeters, voltmeters, ohmmeters, automotive instruments. All use **galvanometers** or **meters**.

Important arguments: Current for full-scale deflection  $I_{fs}$ ; Resistance of the coil  $R_c$

## Ammeters

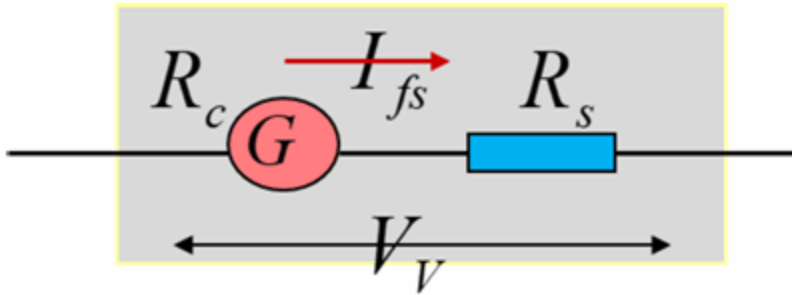
Connect the galvanometers in **parallel** with a shunt resistor  $R_{sh}$ .



$$I_{fs}R_c = (I_a - I_{fs})R_{sh}$$

## Voltmeters

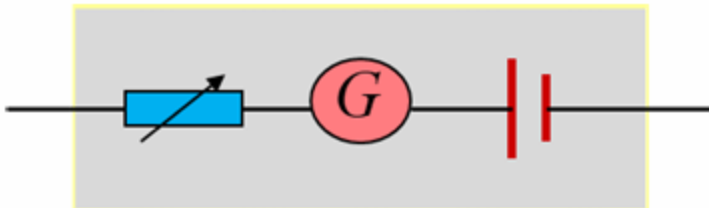
Connect the galvanometers in **series** with voltage divider resistors  $R_s$ .



$$V_V = I_{fs}(R_c + R_s)$$

## Ohmmeters

Consist of a galvanometer, a variable resistor and a power supply connected in series.



The reading direction of an ohmmeter is **opposite** to that of an Ammeter or Voltmeter.

## Resistance-Capacitance Circuit

Charging a capacitor

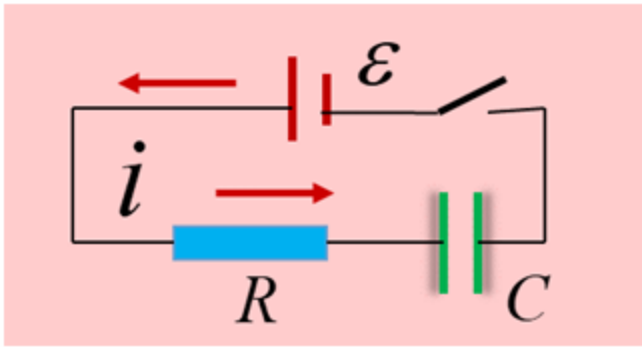
$$\epsilon - iR - q/C = 0$$

where  $i = dq/dt$ . Thus we have

$$q = Q_f(1 - e^{-t/\tau})$$

$$i = I_0 e^{-t/\tau}$$

where  $Q_f = C\epsilon$  is final charge amount,  $I_0 = \frac{\epsilon}{R}$  is initial current.  $\tau = RC$  is time constant, deciding the speed of charging.



Half of the energy is stored in the capacitor and the other half is consumed by the resistor.

Discharging a capacitor:

$$0 - iR - \frac{q}{C} = 0$$

where  $i = dq/dt$ . Thus we have

$$q = Q_0 e^{-t/\tau}$$

$$i = -I_0 e^{-t/\tau}$$

where  $Q_0$  is initial charge amount, the negative sign of  $i$  express the direction is opposite to the charging.  $\tau = RC$ .